Phase 2 - Grade 3 - Probability and Statistics

Key Learning Points -- Basic Probability

Probability measures how likely something is, as a number from 0 (impossible) to 1 (certain). Formula: P(event) = favorable outcomes / total outcomes. If every possible outcome in a situation is listed (nothing left out, nothing overlapping), their probabilities always sum to exactly 1 -- a useful sanity check.

OR rule (mutually exclusive events)

P(A OR B) = P(A) + P(B) -- add probabilities when two events can't both happen at once.

AND rule (independent events)

P(A AND B) = P(A) × P(B) -- multiply probabilities when two events have zero influence on each other. Mechanism: independent events' outcome-combinations multiply (e.g. 2 coin flips = 2×2=4 total combinations), and probability is just favorable/total within that combined count.

Independent vs. dependent events

Test: compare P(Y|X) to P(Y). If equal, independent (knowing X changed nothing). If different, dependent (knowing X shifted the probability of Y).

Key Learning Points -- Conditional Probability

Conditional probability asks: given that X already happened, what's the probability of Y? Mechanism: shrink your denominator down to ONLY the group where X is true, then ask what fraction of that narrowed group also satisfies Y.

P(Y|X) notation

'Probability of Y, given X.' The bar means 'given', not division.

Conditional probability formula

P(Y|X) = P(Y and X) / P(X).

P(A and B) for dependent events

No shortcut multiplication exists -- must come from actual joint data/counts, or from P(A|B) × P(B) (the universal, always-true version of the AND formula, which reduces to simple multiplication only in the special case of independence).

Key Learning Points -- Bayes' Theorem

Bayes' Theorem lets you flip a conditional probability around: if you know P(B|A) but want P(A|B), Bayes' Theorem gets you there. It is NOT a separate idea -- it's the conditional probability formula, rearranged algebraically, because both P(A|B) and P(B|A) share the same numerator P(A and B).

Analogy that made it click: a medical test example. Knowing 'test accuracy given disease' (P(positive|disease)) doesn't directly tell you 'probability of disease given a positive result' (P(disease|positive)) -- that reversal is exactly what Bayes' Theorem solves, and the answer can be surprisingly different from the test's stated accuracy (the 'base rate' effect).

Bayes' Theorem formula

P(A|B) = [P(B|A) × P(A)] / P(B).

Law of Total Probability

Used to find an overall probability (like P(positive)) by splitting the population into every sub-group and summing each group's contribution: P(positive) = P(disease)×P(positive|disease) + P(NOT disease)×P(positive|NOT disease).

Base rate problem

When the thing being tested for is RARE, even a small false-positive rate applied to the huge 'doesn't have it' group can produce more false alarms in raw numbers than true positives from the tiny 'has it' group -- so P(disease|positive) can be much lower than the test's stated accuracy would suggest.

Key Learning Points -- Distributions, Mean, Variance

A **distribution** is every possible outcome paired with its probability (e.g. a bag of marbles, or a die's six faces each at 1/6). **Mean (expected value)** is the long-run average if the random process repeated many times -- not necessarily a value that can actually occur. **Variance** measures how spread out outcomes are around the mean; squaring the deviation makes all terms positive and penalizes bigger deviations more.

Mean formula

Mean = ∑(value × its probability).

Variance formula

Variance = ∑[(value - mean)^2 × its probability].

Clarified Doubts

My doubt: Can you explain how to find probability in general, before I attempt it?

Clarification: Count total possible outcomes (denominator), count how many of those match what's being asked about (numerator), divide. If an outcome type repeats (e.g. 2 red marbles instead of 1), each repeat counts separately toward the numerator.

My doubt: Can you explain the AND rule more -- why does it involve multiplication?

Clarification: Listing out every actual combination (e.g. 2 coin flips -> HH, HT, TH, TT = 4 total combinations) shows multiplication is really just counting how combinations multiply (2×2=4), then checking how many of those combinations match what's wanted (only 1 of 4 is HH) -- which equals the same result as multiplying the individual probabilities (0.5×0.5=0.25).

My doubt: Can you explain where the 9/12 comes from in a conditional probability example?

Clarification: Applying 'given cricket' shrinks the entire population down to ONLY the 12 cricket players (the 8 non-cricket students become irrelevant). Within that narrowed group of 12, count how many also play badminton (9). The conditional probability is just that narrowed count over the narrowed total: 9/12.

My doubt: Does P(A and B) = P(A) × P(B) always apply?

Clarification: No -- only when A and B are INDEPENDENT. For dependent events, that shortcut gives a wrong answer; instead use actual joint counts/data, or the universal formula P(A and B) = P(A|B) × P(B), which works for both dependent and independent events (and simplifies to the multiplication shortcut only

when independent).

My doubt: How do I tell if two events are dependent or independent, and why do the formulas differ?

Clarification: Compare P(Y|X) to the unconditional P(Y): equal means independent, different means dependent. The formulas differ because independent events can be calculated separately and simply multiplied, while dependent events are tangled together and need actual joint data (or the conditional-probability-based formula) rather than a direct multiplication shortcut.

My doubt: Why do we rearrange the second conditional-probability equation and substitute it into the first, when deriving Bayes' Theorem?

Clarification: The formula for P(A|B) needs P(A and B), which isn't directly known. The second equation gives an alternate expression for that exact same P(A and B), built from quantities that ARE known (P(B|A) and P(A)). Since both expressions equal the same real-world quantity, one can be substituted for the other, converting an unmeasurable term into a measurable one.

My doubt: Can you walk through the disease/positive-test example step by step -- what's given, what's asked, and how does the formula substitute?

Clarification: Given: P(disease)=P(A), P(positive|disease)=P(B|A). Asked: P(disease|positive)=P(A|B) -- the REVERSED direction from what's given. Substituting into P(A|B)=[P(B|A)×P(A)]/P(B) requires one more missing piece, P(B)=P(positive) overall, found via the Law of Total Probability (summing true-positive and false-positive contributions across the whole population).

My doubt: Can you explain the base-rate question (why a 95%-accurate test still gives a low P(disease|positive)) in more detail?

Clarification: Breaking the positive-test population into its two sources (true positives from the tiny sick group vs. false positives from the huge healthy group) shows the false-positive count can be several times larger than the true-positive count, purely because the healthy group is so much bigger -- even a small error rate applied to a huge group outweighs a high accuracy rate applied to a tiny group.

Examples

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# Basic probability
Bag: 3 red, 2 blue, 5 green (10 total)
P(red)=0.3, P(blue)=0.2, P(green)=0.5  -- sums to 1.0

# OR rule
P(red OR blue) = 0.3 + 0.2 = 0.5

# AND rule (independent)
2 coin flips -> HH,HT,TH,TT (4 outcomes)
P(heads AND heads) = 0.5 * 0.5 = 0.25  (matches 1/4 from direct count)

# Conditional probability
Classroom: 20 students, 12 play cricket (9 also badminton), 8 don't (2 play badminton)
P(badminton|cricket) = 9/12 = 0.75
P(badminton|NOT cricket) = 2/8 = 0.25
Overall P(badminton) = 11/20 = 0.55  -- differs from 0.75, proving dependence

# Bayes' Theorem
P(disease)=0.01, P(positive|disease)=0.9, P(positive|NOT disease)=0.05
P(positive) = 0.01*0.9 + 0.99*0.05 = 0.009 + 0.0495 = 0.0585
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P(disease|positive) = (0.9*0.01)/0.0585 = 0.009/0.0585 = 0.1538

# Mean & Variance
Spinner: 2 (p=0.5), 8 (p=0.5)
Mean = 2*0.5 + 8*0.5 = 5
Variance = (2-5)^2*0.5 + (8-5)^2*0.5 = 4.5+4.5 = 9

Fair die (1-6, each p=1/6):
Mean = 21/6 = 3.5
Variance = sum[(value-3.5)^2 * 1/6] for value=1..6 = 2.9167
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Quiz Q&A;

Q1. A bag has 4 yellow, 3 purple, 3 orange (10 total). P(yellow)? P(purple OR orange)?

A: P(yellow) = 4/10 = 0.4. P(purple OR orange) = 3/10 + 3/10 = 0.6.

Q2. Roll two fair dice. P(both dice show 6)? Independent or dependent?

A: Independent. P(die shows 6) = 1/6 for each die. P(both show 6) = (1/6)×(1/6) = 1/36 ≈ 0.0278.

Q3. 50 people: 30 coffee (24 also tea), 20 non-coffee (5 tea). P(tea|coffee)? P(tea|NOT coffee)? Independent?

A: P(tea|coffee)=24/30=0.8. P(tea|NOT coffee)=5/20=0.25. Dependent -- these differ sharply from each other and from the overall P(tea)=29/50=0.58, showing coffee-drinking shifts the probability of tea-drinking.

Q4. Disease affects 2%. Test: P(positive|disease)=0.95, P(positive|NOT disease)=0.10. Find P(disease|positive) using Bayes' Theorem.

A: P(positive AND disease) = 0.02×0.95 = 0.019. P(positive AND NOT disease) = 0.98×0.10 = 0.098. P(positive) = 0.019+0.098 = 0.117. P(disease|positive) = 0.019/0.117 ≈ 0.1623.

Q5. A spinner lands on 3 (p=0.4) or 10 (p=0.6). Find the mean.

A: Mean = (3×0.4) + (10×0.6) = 1.2 + 6 = 7.2.

Q6. Why does a 95%-accurate test still give a surprisingly low P(disease|positive) when the disease is rare? What's the underlying cause?

A: Because the disease's base rate is extremely low, the healthy population is overwhelmingly larger than the sick population. Even a small false-positive rate applied to that massive healthy group generates more false alarms in raw numbers than the true positives generated by the test's high accuracy applied to the tiny sick group. The flood of false alarms swamps the few true positives, so a positive result is still more likely to be a false alarm than a real detection.